PORTALDOT HACKER HOUSE 2026 · MODULAR CLOUD & MIDDLEWARE

THE DOCS SAY `transfer`. V3 HAS NEVER HEARD OF IT.

Plumbline checks each call Portaldot documents against the runtime's own metadata, and proves calls with real transactions read back from their blocks.

DOCS → V3 · SPEC 3002

PORTALDOT-V3 · BOOKED 2026-10-19 UTC

STA.	CALL, AS DOCUMENTED	ON V3	RECEIPT
1	BALANCES ~~transfer~~ *V3 has transfer_allow_death, transfer_keep_alive ↓*	**ABSENT**	no such call
2	BALANCES transfer_all	**UNPROVEN**	no tx yet
3	BALANCES transfer_keep_alive	**PROVEN** ✓	106425-1
4	ELECTIONPROVIDERMULTIPHASE ~~governance_fallback~~ *docs: 2 args · V3: none*	**ARG_DRIFT**	2 args → 0
5	SYSTEM remark_with_event	**PROVEN** ✓	106424-1
6	UTILITY batch_all	**PROVEN** ✓	106426-1

```
$ plumbline check examples/mainnet-dapp --target v3   # evidence/check_v3.txt
summary: 9 findings in 9 files scanned · ARGS-UNCHECKED 2, UNPROVEN 5, COMPAT-UNMEASURED 1, UNSCANNED 1 · blocking 0 · exit 0
```

DOCS AND CHAIN DISAGREE

A builder copies `Balances.transfer` from the Portaldot docs. On the V3 testnet that call does not exist, and the page does not say which runtime it describes.

Without a ledger, one way to find out is to submit a transaction and read the error. The next runtime upgrade moves the line again.

	WHAT THE SOURCES SAID	WHAT V3 REPORTS
ENDPOINT	~~`testnetv3-node.feso-apps.xyz`~~ *V3 testnet guide §1.1 · no DNS on 2026-10-14*	`test-node-v3.portaldot.dev`
SPEC	~~`portaldotV3 / 3000`~~ *a third-party README, not re-verified*	`portaldot-v3 / 3002`
SYMBOL	~~`tPOTv3`~~ *V3 testnet guide §1.1, §3.1*	`tPOT`
PALLETS	~~70~~ *the same README, not re-verified*	69

79 → 2

Of the **196** calls in Portaldot's mainnet docs, **79** disagree with mainnet spec 1002 (64 absent or renamed, 15 with other arguments). **2** disagree with V3 spec 3002: `Balances.transfer` and `governance_fallback`. The pages name no spec.

CALL, VERDICT, PROOF

Sta. 01

DOCUMENTED CALL

22 mainnet docs pages give 196 calls. The V3 testnet guide at `d671a32` gives 46 `compose_call` sites and 12 FAQ modules, read by AST and text, never executed.

`plumbline docs · guide`

→ Sta. 02

VERDICT

Each entry is joined to one runtime's metadata and gets exactly one verdict, stamped with profile, spec version and metadata hash. Same inputs, byte-identical ledger.

`plumbline plumb`

→ Sta. 03

PROOF

A canary extrinsic is signed and sent, its receipt kept, then re-read from its block by hash in a new process. PROVEN needs that receipt.

`plumbline canary · readback`

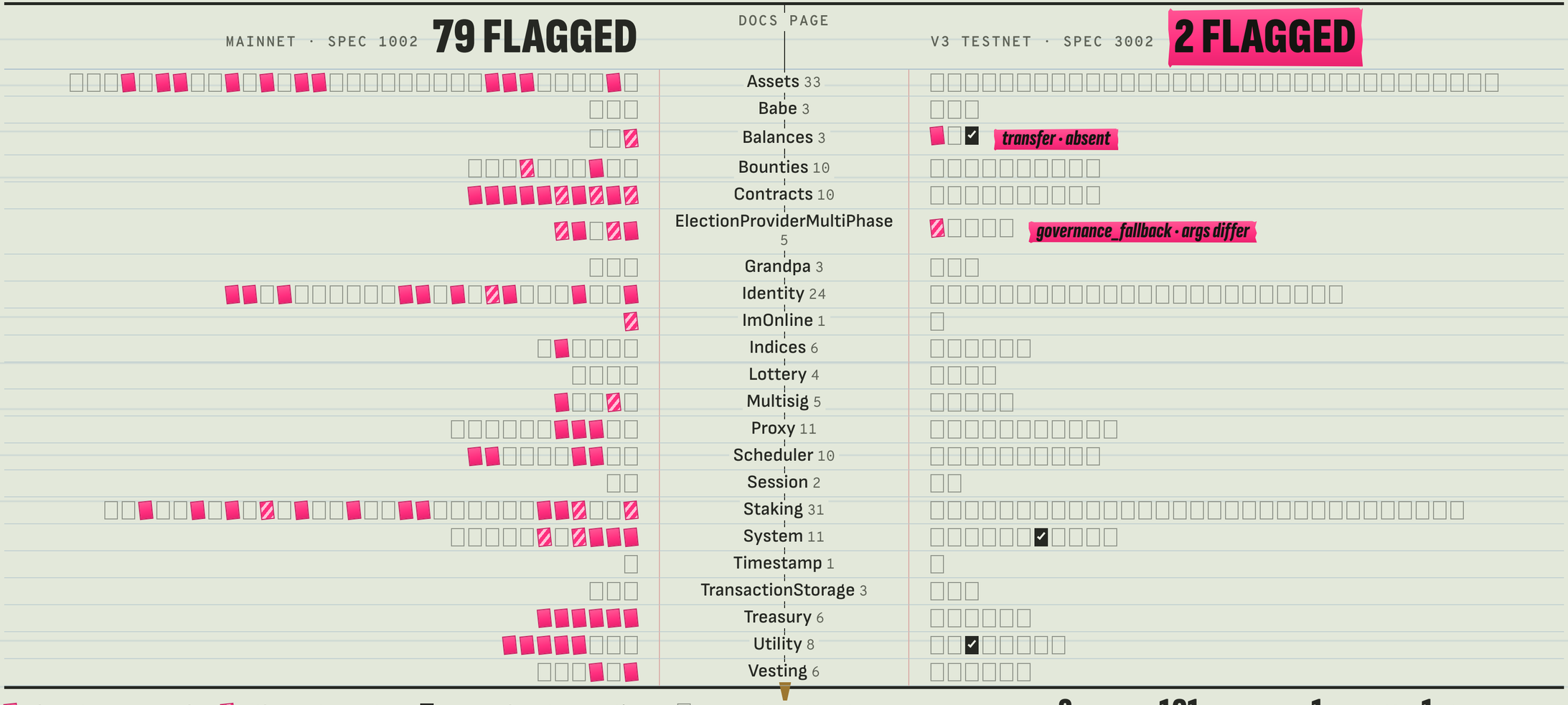
PROVEN | UNPROVEN | ARGS-UNCHECKED | ARG_DRIFT | DRIFTED | ABSENT

verdicts from plumbline/verdicts.py

Beside the flow: `plumbline diff A B` lists what moved between two runtimes; `plumbline check <dApp> --target v3` classifies a dApp's call sites against one runtime and exits 0 / 1 / 2.

SAME 196 CALLS. TWO RUNTIMES.

Every call in Portaldot's mainnet docs, booked by docs page. Left: hung against mainnet spec 1002. Right: the same calls against the V3 testnet, spec 3002, on 2026-10-19.



absent or renamed · other arguments · proven by a transaction · present, no tx yet

3 proven · **191** unproven · **1** arg drift · **1** absent on V3

THREE TRANSACTIONS, READ BACK

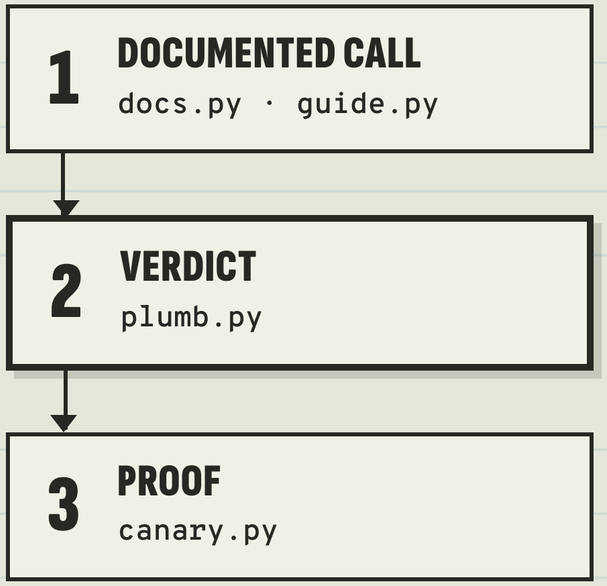
Portaldot V3 testnet · portaldot-v3 / spec 3002 · signer 5EhP3KWt…RUzDsb · canary run 20261019T182014Z

EXTRINSIC	CALL	TX HASH	READ BACK	NODE CONSOLE
106424-1	System.remark_with_event	0xf544305710…c941e6f5	**MATCH** ✓	block 106424 ↗
106425-1	Balances.transfer_keep_alive	0xa7effe5d5a…ff91d90b	**MATCH** ✓	block 106425 ↗
106426-1	Utility.batch_all	0x25e0352284…0f5dfaa8	**MATCH** ✓	block 106426 ↗

EVIDENCE/READBACK_V3.TXT · VERBATIM — NEW PROCESS · NEW CONNECTION

```
# plumbline readback evidence/canary_v3.json — profile v3, run 2026-10-19T18:24:33Z, new process, new connection
   106424-1 System.remark_with_event         decoded System.remark_with_event  block 106424  success True  MATCH
   106425-1 Balances.transfer_keep_alive     decoded Balances.transfer_keep_alive  block 106425  success True  MATCH
   106426-1 Utility.batch_all                decoded Utility.batch_all  block 106426  success True  MATCH
profile v3 — Portaldot V3 testnet: 3/3 receipts match the chain
```

SOURCES, ENGINE, EVIDENCE



← mainnet docs: 22 pages, 196 calls

← V3 testnet guide d671a32: parsed, never run

← metadata.py: V3 spec 3002, read-only, one pinned block

→ evidence/ledger_v3.json: 549 rows, one verdict each

→ 3 transactions on V3, read back 3/3 by block hash

→ docs/PROOF.md

↑ receipts feed step 2 (--receipts canary_v3.json): PROVEN needs a receipt.

Beside the flow: diff.py, two runtimes, 516 rows · check.py, a dApp against one runtime, exit 0 / 1 / 2

UNTRUSTED INPUTS

The V3 guide is cloned outside the repo, parsed by AST and text, never run.

EVERY STATION WRITES A FILE

so every figure in this deck has a committed file behind it.

THE FULL DRAWING

docs/architecture.png adds the inputs, check.py, diff.py and the chains; its dashed box is the planned Ledger contract.

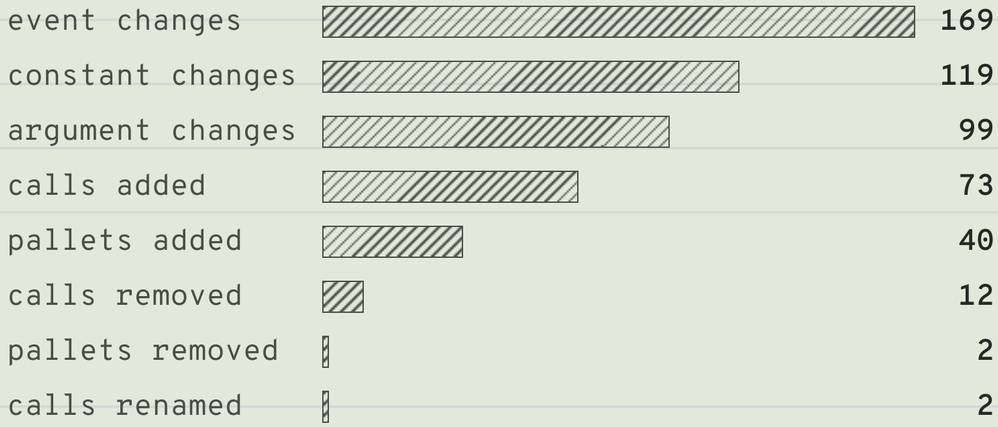
A VERDICT NEEDS A RECEIPT

DOCS PAGES	what was designed; the pages name no runtime spec
NODE-CONSOLE	Portaldot's explorer: what exists at one block, not what the docs promised
METADATA DIFF	what moved between two runtimes, not whether the docs match either one
ONE TEST TX	that one call worked once, on one runtime

Plumbline joins them: each docs entry against the runtime's own metadata, one verdict stamped with the spec, PROVEN only with a receipt, re-run on the next runtime.

WHAT MOVED, MAINNET SPEC 1002 → V3 SPEC 3002

516 rows from `plumbline diff` · evidence/diff_1002_v3.json



5 → 0 blocking findings for the same example dApp under `plumbline check`: on spec 1002, then on V3

BETWEEN PORTALDOT'S DOCS AND ITS RUNTIME

DOCUMENTED

WHAT BUILDERS READ

`portaldot-dev.readthedocs.io` extrinsic pages, and the third-party V3 testnet guide (`d671a32`) with its FAQ

PLUMBLINE

THE MIDDLEWARE

PLUMB • DIFF • CHECK • CANARY

through the SDK Portaldot's docs point to, `substrate-interface 1.8.1`. The canary uses `compose_call · create_signed_extrinsic · submit_extrinsic · get_block · get_events`; the read-only commands use RPC calls such as `state_getMetadata`.

RUNTIMES

WHAT ACTUALLY RUNS

V3 testnet `portaldot-v3 / 3002` · mainnet spec 1002, read-only: `state_getMetadata · state_getRuntimeVersion · state_getStorageHash(:code)`

PINNED TO THE REAL V3

The `v3` profile checks spec name `portaldot-v3`, genesis `0xd4f9acbc…5283bd` and spec 3002, and fails loudly on a mismatch.

RE-RUN PER UPGRADE

The spec 1002 baseline and the spec 3002 sheet come from the same commands; the next runtime is the next run.

WHO RUNS IT, WHO PAYS

WHO	MOMENT OF PAIN	WHAT THEY GET	STATUS
HACKER HOUSE BUILDERS NOW · THE FIRST AUDIENCE	a documented call fails on V3	`plumbline check <dApp> --target v3`; exit 1 on blocking findings, so it can gate CI	TODAY
TEAMS MOVING 1002 → V3	not knowing what breaks on V3	the 516-row runtime diff, and `check` run on their repo	TODAY
PORTALDOT DOCS & DEVREL TARGET CUSTOMER	re-verifying every page per runtime by hand	the ledger as docs CI: a “verified on spec N” line per reference entry	PLANNED
WALLETS & INTEGRATORS AFTER THE EVENT	an upgrade changes call shapes	a diff and a canary re-run per upgrade; always-on alerts as the paid tier	PLANNED

MODEL

Docs CI as an ecosystem-funded public good, plus paid upgrade alerts for integrators. Portaldot is the first chain; any Substrate chain whose docs trail its runtime has the same gap.

HONEST LIMIT

No revenue and no external users yet. The first two rows work today on the public repo; the last two are the plan.

RE-RUN EVERY NUMBER HERE

549 verdict rows on V3, spec 3002: 297 documented entries + 252 runtime calls no source names; byte-identical on re-run

3 / 3 V3 transactions read back from their blocks

263 offline tests passed in the stored run (evidence/pytest_checkin.txt, at c484fcb) · 116 commits from 2026-10-14 to build 2460a0d

planned a canary per documented call (3 of 196 PROVEN today)

planned the Plumbline Ledger contract (Revive) on V3

open gap `check` does not detect TYPE-TRAP yet; runtime APIs (ReviveApi) are not checked yet

```
$ pip install -e '.[dev]' && pytest -q
$ plumbline check examples/mainnet-dapp --target v3
# … · blocking 0 · exit 0
```

plumbline.edycu.dev

Scan: the landing page

github.com/edycutjong/portaldot-hackathon-2026-plumbline-edycu

proof: docs/PROOF.md · docs/BASELINE-V3.md

video: youtu.be/edvFu0dUZNk

AI assistance: Claude Code drafted most of the code (the author reviews it; see the README's AI disclosure) and drafted this deck from the repo's evidence.